DATAPULSE AUTONOMOUS

AI-DRIVEN BIG DATA INTELLIGENCE FOR SELF-MONITORING, DRIFT DETECTION & PIPELINE RISK

THE PROBLEM	THE PROPOSED SOLUTION
Large data pipelines can run successfully while the data itself becomes incomplete, inconsistent, anomalous or statistically different from historical behaviour.	A scalable platform that profiles every data batch, detects abnormal behaviour, predicts risk, explains incidents with AI and performs safe, verified remediation with minimal human intervention.

HOW IT WORKS

1. INGEST Multi-GB NYC Taxi TLC data	→	2. PROCESS HDFS / MapReduce / Spark	→	3. PROFILE Quality + drift + pipeline metrics	→	4. PREDICT ML risk model	→	5. ACT AI diagnosis + safe remediation

BIG DATA ENGINEERING

- **NYC TLC Trip Records** across multiple months/years
- Distributed storage using HDFS / object storage
- Custom MapReduce jobs for aggregation & profiling
- Spark for scalable transformations
- Partitioned Parquet analytical datasets

ANALYTICAL ENGINEERING

- Batch-level data profiles
- Data-quality metrics
- Distribution and drift metrics
- Pipeline performance metrics
- Reusable analytical tables for downstream analysis

INTELLIGENCE LAYER

- Schema-drift detection
- Volume and completeness anomalies
- PSI / KS / statistical shift analysis
- ML-based future risk estimation
- Time-aware model evaluation

AI + AUTONOMATION

- Ollama-based local AI diagnostician
- Allowlisted remediation tools
- Policy-controlled autonomous actions
- Post-action verification
- Human escalation for uncertain/high-risk cases

AUTONOMOUS OPERATIONS

OBSERVE	→	DETECT	→	PREDICT	→	DIAGNOSE	→	REMEDIATE	→	VERIFY	→	ESCALATE

EXAMPLE: WHAT HAPPENS WHEN DATA GOES WRONG?

Injected problem	System response	Autonomous outcome
A new batch suddenly contains 20% missing fare values and a major shift in fare distribution.	Detect → compare with baseline → calculate drift → create incident → ML risk assessment → Ollama diagnosis.	Quarantine affected records → rebuild clean analytical batch → run quality checks → verify recovery → close incident or escalate.

PROJECT OBJECTIVES	EXPECTED OUTPUT
1. Demonstrate distributed big-data processing. 2. Build analytical and data-quality layers. 3. Detect and quantify drift. 4. Predict future data/pipeline risk. 5. Demonstrate AI-assisted autonomous operations.	A reproducible multi-GB data platform with MapReduce/Spark processing, analytical datasets, ML risk scoring, an Ollama AI control plane, controlled remediation, verification and a dashboard showing system health and autonomy metrics.

DATA SOURCE: NYC TAXI & LIMOUSINE COMMISSION (TLC) TRIP RECORD DATA

TAGLINE: OBSERVE • UNDERSTAND • ACT • VERIFY